
ALTEGRAD Kaggle Challenge: Molecular Graph Captioning

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Abstract

This report presents two methodologies for the Molecular Graph Captioning task within the ALTEGRAD Kaggle challenge [1]. We compare a **Contrastive Approach** (SciBERT aligned with a GNN) and a **Generative Approach** (Galactica LLM conditioned on graph features). Our findings indicate that the generative approach is highly promising, yielding superior results compared to the contrastive approach for an equivalent training duration. However, while the contrastive model was the one trained for the longest period in our experiments, the generative approach would require more extensive training and a larger dataset to become truly decisive. Furthermore, we demonstrate that adding a discriminator to the contrastive alignment provides no significant performance gain. The implementations for both approaches are available in the Code directory as two separate notebooks, one dedicated to each method.

1 Introduction

The task of molecular captioning requires bridging the gap between structural chemistry and natural language. In the context of the Kaggle challenge, models must generate descriptive captions for molecules represented as graphs. We explore two distinct paradigms: a latent-space alignment via contrastive learning and a direct conditional generation using a scientific large language model (LLM).

2 Methodology

2.1 Contrastive Approach

The contrastive framework aims to learn a shared embedding space where a molecular graph G and its description T are close if they represent the same entity. See Figure 1 for an overview of the pipeline.

Graph Encoder with Node and Edge Features. The graph encoder relies on a deep structural architecture centered around three **GINEConv** layers and a global additive pooling mechanism. This design is specifically chosen for its theoretical expressiveness, as GIN operators are maximally discriminative among GNNs. By integrating edge features, the GINE variant ensures that distinct molecular topologies and bond characteristics are mapped to unique embeddings.

Initially, the model processes discrete chemical information through parallel **embedding** look-up tables for node features (including atomic number, hybridization, and valency) and edge attributes (bond type, stereochemistry, and conjugation). These representations are summed to map multidimensional chemical data into a dense, unified latent space.

The core processing is performed by three successive **GINEConv** layers, explicitly selected to integrate bond-level information (edges) into the atom-level updates (nodes) during message passing.

Each convolution incorporates an internal multi-layer perceptron (MLP) to model complex topological relationships and refine the local chemical context, followed by a ReLU activation.

To aggregate these local features into a global representation, the model employs a **Global Add** pooling layer (`global_add_pool`), which sums the node embeddings across the entire molecular graph. This resulting vector is then processed through a linear projection head to align the dimensionality with the text embeddings.

LoRA-Enhanced SciBERT. For textual representation, we employ **SciBERT**, a language model pre-trained on a vast corpus of scientific literature, augmented with **LoRA** (Low-Rank Adaptation). Unlike generic BERT models, SciBERT is specifically optimized for scientific discourse, allowing for a more precise tokenization and semantic understanding of complex chemical nomenclature and molecular descriptions. This domain-specific initialization provides a robust foundation, while LoRA ensures parameter-efficient fine-tuning. The representation is derived from the `last_hidden_state` of the `[CLS]` token and projected into the shared latent space.

Loss. The model optimization is driven by a symmetric contrastive loss function, implemented as a temperature-scaled Cross-Entropy loss over the similarity matrix. Let $h_G \in \mathbb{R}^d$ and $h_T \in \mathbb{R}^d$ denote the L_2 -normalized embeddings of the molecular graph and its corresponding text description, respectively. For a batch of N pairs $\{(G_i, T_i)\}_{i=1}^N$, we compute the similarity matrix $S \in \mathbb{R}^{N \times N}$ where $S_{i,j} = (h_{G_i} \cdot h_{T_j}^T) \cdot \exp(\tau)$.

The training objective aims to maximize the similarity of the diagonal elements (matched pairs) while minimizing off-diagonal elements (mismatched pairs). This is achieved by minimizing the symmetric loss $\mathcal{L}_{total} = \frac{1}{2}(\mathcal{L}_{G \rightarrow T} + \mathcal{L}_{T \rightarrow G})$, where:

$$\mathcal{L}_{G \rightarrow T} = -\frac{1}{N} \sum_{i=1}^N \log \frac{\exp(S_{i,i})}{\sum_{j=1}^N \exp(S_{i,j})} \quad \text{and} \quad \mathcal{L}_{T \rightarrow G} = -\frac{1}{N} \sum_{i=1}^N \log \frac{\exp(S_{i,i})}{\sum_{j=1}^N \exp(S_{j,i})} \quad (1)$$

Here, τ is a learnable temperature parameter (clamped at 4.6 to prevent training instability) that scales the logits. Intuitively, this loss forces the graph and text representations of the same molecule to be projected close to each other in the joint latent space.

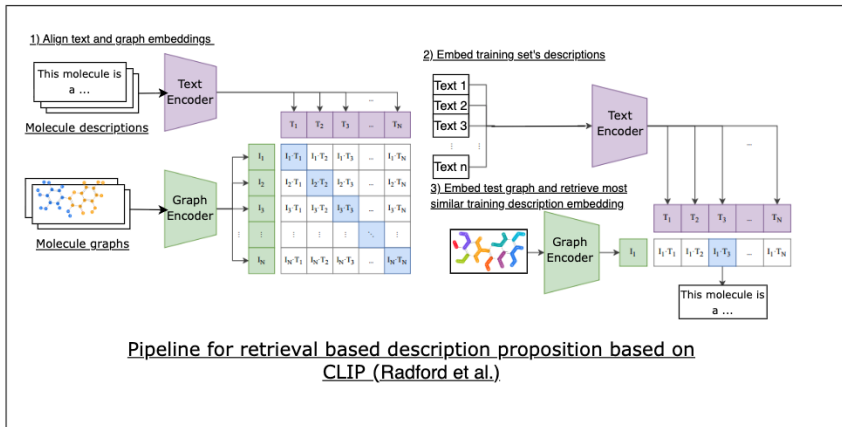


Figure 1: Contrastive model training and retrieval Pipeline. Inspired by a figure from [3].

In addition to this primary objective, we implemented a variant incorporating an auxiliary adversarial **Discriminator**. This binary classifier was trained to distinguish between true pairs (h_{G_i}, h_{T_i}) and negative samples constructed by shuffling the batch. The hypothesis was that the discriminator would act as a regularizer, pushing the encoder to generate multimodal features that are indistinguishable in terms of distribution, thereby refining the alignment beyond simple instance-level retrieval. The training follows an alternating adversarial scheme. First, the **Discriminator** is optimized to correctly classify the source modality (Graph vs. Text) of the embeddings, while the encoders are frozen. Subsequently, the **Encoders** are updated to simultaneously minimize the contrastive alignment loss and maximize the discriminator’s error (fooling objective), thereby enforcing domain invariance between the two modalities.

2.2 Generative Approach

The second methodology treats captioning as a sequence generation problem $P(T|G)$. See Figure 2 for an overview of the pipeline.

Galactica Backbone. We utilize **Galactica** (125M version) [4], an LLM specifically designed for scientific tasks. Its extensive pre-training on scientific documents provides a critical advantage for generating specialized chemical descriptions compared to general-purpose models. Furthermore, selecting this compact version allows for efficient fine-tuning within strict memory constraints, making it feasible to train on available hardware while retaining strong domain capabilities.

MolCA Inspiration. Inspired by the **MolCA** architecture [2], we implement a comprehensive cross-modal generation pipeline designed to condition an LLM on molecular structures. The architecture consists of three main components: a graph encoder, a multi-token projector, and the Galactica LLM.

Graph Encoding Strategy. We initially utilized a standard graph encoder based on GINEConv layers and global mean pooling. However, to better capture the structural nuances, we transitioned to an **Attentive Graph Encoder**. This improved module employs three layers of **GATv2Conv** (Graph Attention Networks v2), which dynamically weigh the importance of neighboring atoms during message passing. Crucially, the final aggregation is performed by an **AttentionalAggregation** layer (controlled by a gating network `gate_nn`), replacing simple averaging. We hypothesize that this attention mechanism allows the model to "focus" on specific pharmacophores or functional groups that are pivotal for the textual description, thereby generating a more informative graph embedding h_G .

Cross-Modal Projection. To bridge the modality gap, the fixed-size graph embedding h_G is processed by a **Multi-Token Projector**. Unlike simple linear layers that project to a single vector, this module transforms the graph embedding into a sequence of $N = 8$ continuous tokens. These "soft tokens" share the same dimensionality as the LLM’s word embeddings, effectively acting as a learnable prefix.

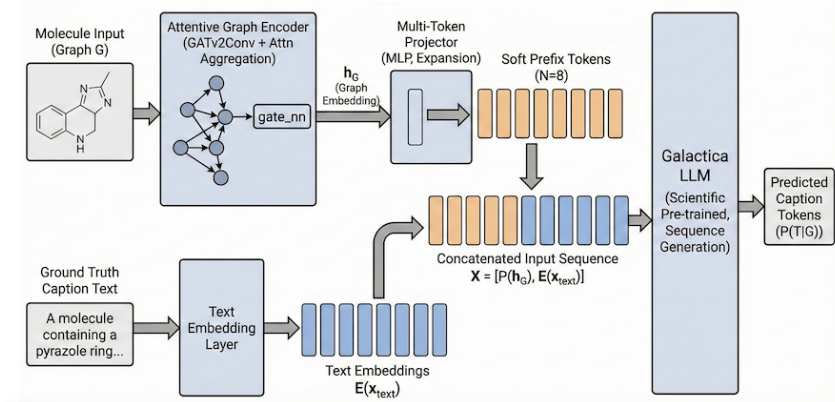


Figure 2: Generative model Pipeline

Training Protocol. The training follows a conditional causal language modeling objective. The input sequence to the LLM is formed by concatenating the projected graph tokens and the text tokens: $X = [P(h_G), E(x_{text})]$. The model is trained end-to-end:

1. We train the **Graph Encoder** and the **Projector** (learning rate 10^{-3}) to align the visual features with the semantic space.
2. We simultaneously fine-tune the full **Galactica LLM** (learning rate 5×10^{-5}) to adapt its scientific knowledge to the specific captioning task.
3. The loss function is a shifted Cross-Entropy Loss that ignores the graph prefix tokens (via `ignore_index=-100`), ensuring the model learns to predict the text based on the graph, without trying to reconstruct the graph tokens themselves.

Inference Pipeline. During inference, the process is autoregressive. The molecule is first encoded and projected into the soft prefix. Generation is initialized with a start token (e.g., ‘<s>’ or ‘[START]’). At each step, the model receives the concatenation of the *fixed* graph prefix and the *growing* sequence of generated text tokens to predict the next token via greedy decoding, until an end-of-sequence token is produced.

3 Experimental Results

3.1 Training Efficiency and Convergence

- **Contrastive Approach:** We trained the contrastive model for **30 epochs**. Fine-tuning SciBERT was highly efficient, requiring approximately **3 minutes per epoch**.
- **Generative Approach:** Due to VRAM constraints, we were limited to the 125M Galactica model. Despite this smaller size, the autoregressive generation process took longer, averaging **10 minutes per epoch**. Consequently, we restricted the training to **10 epochs**. This early stopping was necessitated by the emergence of overfitting trends discussed more in depth later on.

In both approaches we save the model having the best performance on the validation set.

3.2 Performance Analysis

The Discriminator’s Role. Our experiments show that the addition of the **discriminator does not bring significant value**. The contrastive loss alone is powerful enough to align the representations, and the adversarial overhead did not translate into better retrieval or generation metrics.

Kaggle Metrics. When evaluating the performance at an **equivalent number of training epochs** (10), a **clear advantage for the generative model** was observed. Galactica’s ability to construct novel, linguistically coherent sentences based on the graph prompt outperformed the contrastive method’s initial alignment. However, our overall best performance on the Kaggle leaderboard was achieved using the **contrastive approach** after extended training (30 epochs), demonstrating that the retrieval-based alignment benefits significantly from longer optimization.

Table 1: Comparison of Model Performance and Training Efficiency.

Approach	Backbone	Training Epochs	Kaggle Score
Contrastive	SciBERT + GNN	10	0.56
Generative	Galactica (125M)	10	0.60
Contrastive	SciBERT + GNN	30	0.66

To further quantify the alignment quality of the **contrastive framework**, we evaluated its retrieval performance on the validation set using standard ranking metrics: Mean Reciprocal Rank (MRR) and Recall at k ($R@k$). The best model achieved an **MRR of 0.7704**, indicating that the ground-truth text description is, on average, ranked between the first and second position across all queries. The **Recall@1 of 0.6450** demonstrates that the correct description is the top-ranked match in 64.5% of cases, while the **Recall@10 reached 0.9670**. This near-perfect Recall@10 confirms that the joint embedding space effectively clusters relevant modalities, as the correct description is captured within the top ten candidates in over 96% of instances. These results underscore the robustness of the SciBERT-GNN alignment in identifying structural-linguistic correspondences.

Qualitative Analysis. To demonstrate the qualitative capabilities of the **generative pipeline**, we examined the model’s output after 10 training epochs. At this stage, the **MultiTokenProjector** has effectively learned to translate the structural graph features into the continuous embedding space of Galactica. As shown in the example below, the model does not merely generate generic chemical terms. It accurately identifies the specific molecule and its biological function, mirroring the ground truth with high fidelity.

Overfitting of the generative model. While the initial optimization phase showed a steady decline in training loss, clear signs of **overfitting** emerged around the 7th epoch, where the validation loss

began to diverge. This phenomenon is likely a consequence of the Galactica backbone’s high capacity relative to the dataset’s specificity, causing the model to memorize exact linguistic sequences rather than generalizing the fundamental mapping between graph topology and chemical nomenclature.

Qualitative Example (Generative Approach - 10 Epochs):

Reference: *The molecule is tetraanion of ADP-D-ribose 2'-phosphate. It has a role as a human metabolite. It is a purine ribonucleoside 5'-diphosphate, a member of adenosines and a carbohydrate phosphate.*

Prediction: *The molecule is tetraanion of ADP-D-ribose 2'-phosphate. It has a role as a human metabolite. It is a purine ribonucleoside 5'-diphosphate...*

4 Conclusion

In this study, we explored two distinct paradigms for bridging the gap between molecular structures and natural language: a bi-encoder contrastive framework and a generative multi-token projection approach. Our analysis reveals that while the **generative approach is the most promising** due to its linguistic flexibility and ability to synthesize complex pharmacological descriptions, the **contrastive approach** remains more robust under limited computational resources. Given the constraints on training time and data availability, the contrastive model (SciBERT + GNN) achieved our highest performance with a **Kaggle public score of 0.66**. Ultimately, the synergy between specialized graph encoders and large-scale language models represents a powerful frontier for drug discovery and automated chemical reporting.

Several avenues for improvement could further enhance the alignment between chemical and linguistic modalities:

- **Backbone Scaling:** For the generative approach, utilizing larger variants of **Galactica** (e.g., 1.3B or 6.7B parameters) would provide a deeper inherent understanding of pharmacological descriptions. However, it is important to note that such an increase in model capacity would necessitate a **significantly larger training dataset** to mitigate the risk of overfitting observed in our experiments and to ensure robust generalization.
- **Multimodal Hybrid Representations:** Integrating the **SMILES string** representation alongside the molecular graph would allow the model to benefit from both topological (graph) and sequential (string) information.
- **Cross-Attention Mechanisms:** Replacing simple additive pooling or concatenation with **cross-attention** between the graph embeddings and the LLM’s hidden states would allow for a more dynamic and fine-grained interaction between the two modalities.
- **Avoiding Overfitting:** To mitigate the rapid convergence to memorized patterns, future work should implement rigorous regularization strategies combined with graph data augmentation. Techniques such as subgraph masking, node dropping, and increased weight decay would enhance sample diversity and constrain model complexity, thereby improving robustness and generalization on unseen data.
- **Generation-Based Retrieval:** Alternatively, the pipeline could prioritize the generative model to first synthesize a candidate description. This generated text would then act as a query to search the training corpus, retrieving the specific ground-truth description that maximizes textual similarity metrics such as BLEU or ROUGE. This approach grounds the model’s hallucinations by mapping them back to the closest verified real-world example.

References

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